JoSAA allocation audit

Methods, counterfactual results and limitations

STATUS: Project-hosted technical note. This is an implementation and simulation audit, not a new proof of applicant DSIC or an evaluation of real student outcomes.

1. The question

How do an applicant's round decisions interact with vacancies created by other applicants withdrawing? Student-proposing deferred acceptance is already known to satisfy applicant dominant-strategy incentive compatibility (DSIC) under the standard single-round assumptions. This portfolio uses that result as a baseline and studies a simplified multi-round mechanism through paired counterfactuals.

2. Headline checks

Check	Observed result	Interpretation
Static reporting	0 profitable deviations / 1,080 tested	60 applicants, 17 programs; one synthetic cohort.
Freeze versus Float	50 / 50 trials missed an upgrade under Freeze	One selected applicant; identical withdrawal in both worlds.
Dynamic workload	201 matching runs	One target-selection allocation, then 50 pairs x 2 rounds.

3. What the data actually represents

The input file data/raw/sample_josaa_cutoffs.csv is a sample opening/closing-rank table in the repository. It does not include a verified official provenance URL or a year. Candidate preference lists and utilities are synthetic. The reported checks therefore concern constructed markets derived from this sample, not real submitted JoSAA preferences or observed applicant regret.

The current calibration uses OPEN-category rows from round 1. Capacity is approximated from each rank-band width, with a minimum of one seat. Program utilities depend on closing-rank order, creating a strong shared preference structure. The dynamic audit does not test quota-specific clearing rates.

The paired experiment

4. Reproduce the decision timing

The dynamic script constructs 80 synthetic candidates over 17 programs. A round-1 allocation selects the first allocated applicant who did not receive their first choice: applicant_005. Four allocated candidates have higher merit priority. Each trial samples exactly one of those four for withdrawal before round 2.

Both counterfactual worlds use the same initial cohort, program capacities and sampled withdrawal. In the Float world the target retains upgrade eligibility. In the Freeze world the target commits at the end of round 1 and restricts their next-round options to the held seat. The script uses the current engine functions rather than the removed class API.

There are only four possible single-withdrawal sets in this design, sampled repeatedly over 50 trials with seed 1337. The denominator is 50 paired trials for one selected applicant, not 50 different students. All 50 produce a missed upgrade for the frozen target. This conditional result must not be turned into a population-wide percentage.

5. Inspectable outputs

evidence/josaa-static.json records cohort size, tested deviations, profitable deviations and assignment counts. evidence/josaa-dynamic.json additionally records the target, each withdrawal set, both final matches, preference-rank differences and utility differences. Both files contain SHA-256 fingerprints for the scripts, engine and input sample.

6. Checks and implementation detail

The regression suite checks hand-solved allocations, vacancy chains, action filtering and the agreement between a retained student's current_match and the program allocation. A metadata bug for later higher-priority proposals was repaired while verifying these benchmarks. The finite static audit enumerates truncations and two adjacent top-choice swaps per applicant.

A matching pass permits at most S x P distinct proposals. This Python reference uses a list queue and sorts the retained candidate pool on each proposal; queue removal and capacity-dependent sorting add overhead. It is an inspectable reference implementation, not an optimized production allocator.

Reproduction & boundaries

7. Run from the repository root

```
python -m pytest -q
python -m scripts.benchmark_manipulability --json evidence/josaa-static.json
python -m scripts.audit_dynamic_strategies --json evidence/josaa-dynamic.json
```

Python standard-library kernels; pytest for checks. The browser sandbox is a separate JavaScript/Tailwind interface. The static research portfolio uses HTML, CSS, JavaScript ES modules and SVG, delivered with GitHub Pages.

8. What the result does not establish

The sample table is not a complete official seat matrix. Submitted preferences, horizontal and supernumerary rules, fees, risk preferences and alternative admissions are not estimated. The benchmark isolates a two-round, OPEN-category construction; the reservation component has separate tests and is not part of this dynamic estimate.

No Round-4 cutoff, category-specific effect, 50,000-candidate experiment or general regret rate is claimed. The static finite check is consistent with established DSIC theory under the reference assumptions; it neither proves the theorem nor certifies the complete real admissions mechanism.

9. Next extensions

Archive verifiable official inputs and their provenance; diversify preference models; vary target selection and withdrawal rules; compare multiple seeds and cohort sizes; implement and test richer category and timing rules. User studies and independent review would provide different evidence from numerical tests and should be reported separately.

Sources & artifacts

github.com/thetarocks/catalyst-lab
lab.yourcatalyst.in
Primary implementation: core/deferred_acceptance.py; core/josaa_engine.py; core/agents.py.
Benchmark outputs: evidence/josaa-static.json; evidence/josaa-dynamic.json.

The implementation and notes were developed with AI coding assistance. Distinguish understood, personally validated contributions from tool-assisted implementation work when describing the portfolio.